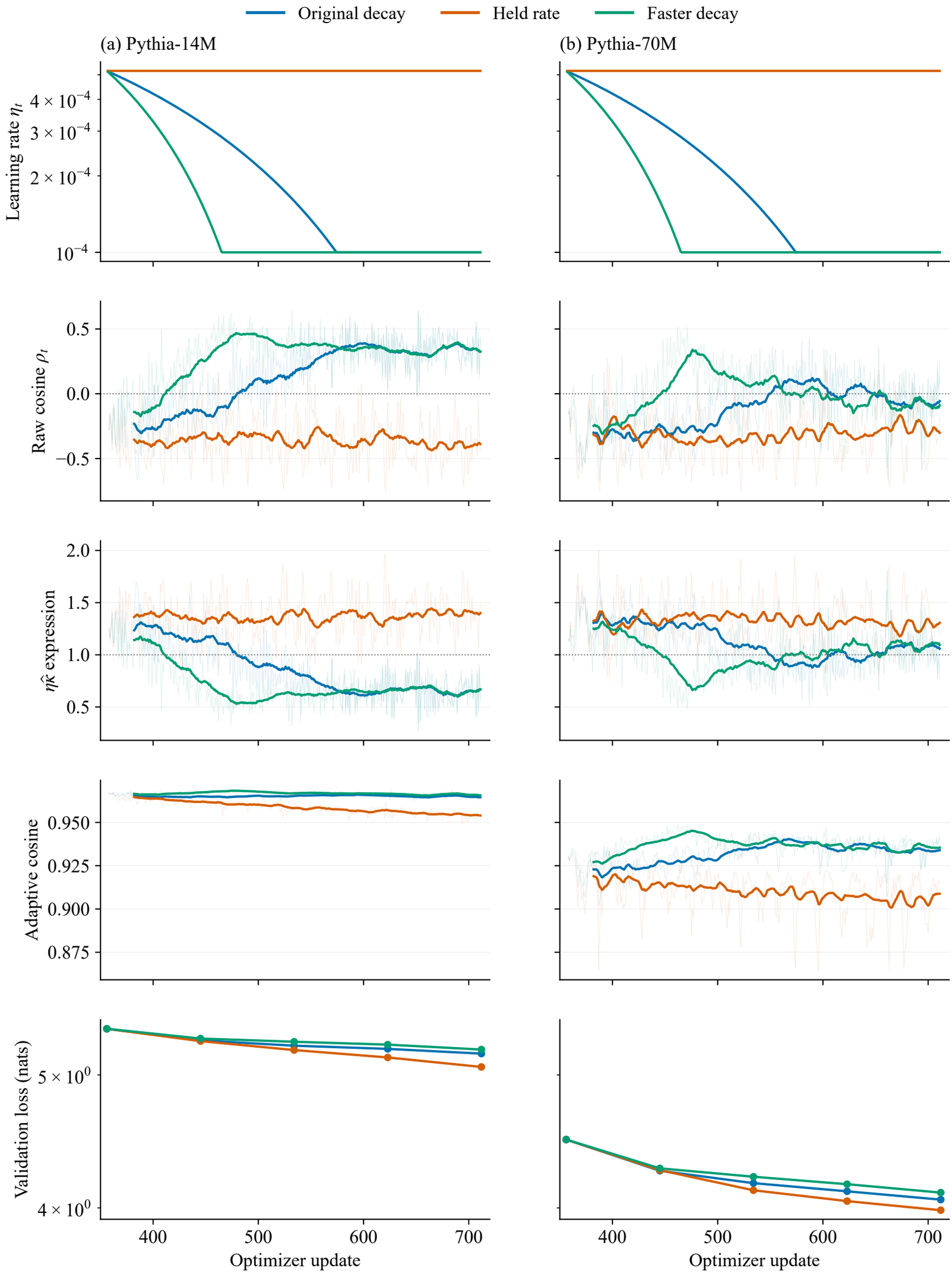

Original decay
Held rate
Faster decay
(a) Pythia-14M
(b) Pythia-70M
Learning rate η_t
4×10^{-4}
3×10^{-4}
2×10^{-4}
10^{-4}
Raw cosine ρ_t
0.5
0.0
−0.5
$\eta\hat{\kappa}$ expression
2.0
1.5
1.0
0.5
Adaptive cosine
0.950
0.925
0.900
0.875
Validation loss (nats)
5×10^{0}
4×10^{0}
400
500
600
700
Optimizer update